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MachineLearning

EXPLORER

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OUTLINE
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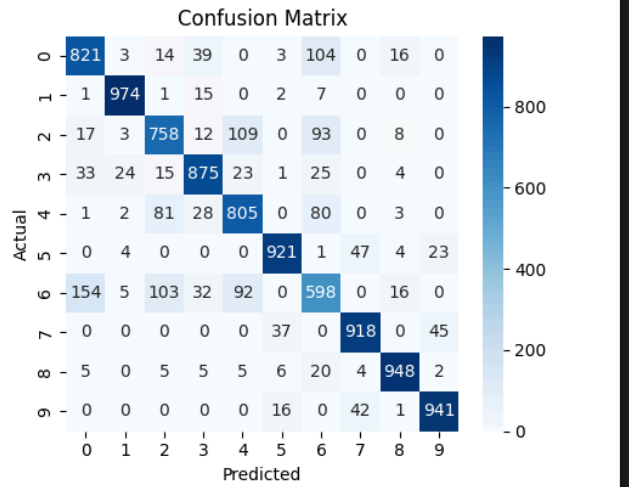
```python
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix

train_df = pd.read_csv("fashion-mnist_train.csv")
test_df = pd.read_csv("fashion-mnist_test.csv")


# Since it's a greyscale image, I just normalized
# the pixel values to be between 0 and 1 by dividing by 255.0.
y_train = train_df.iloc[:, 0].to_numpy()
X_train = train_df.iloc[:, 1:].to_numpy() / 255.0

y_test = test_df.iloc[:, 0].to_numpy()
X_test = test_df.iloc[:, 1:].to_numpy() / 255.0
print("Data loaded successfully.")

model = LogisticRegression(solver='saga', max_iter=1000, tol = 1e-2)
print("Training the model...")
model.fit(X_train, y_train)

print("Model trained successfully.")
y_pred = model.predict(X_test)

acc = accuracy_score(y_test, y_pred)
print("Accuracy:", acc)

print("\nClassification Report:")
print(classification_report(y_test, y_pred))

cm = confusion_matrix(y_test, y_pred)
print("\nConfusion Matrix (Text):")
print(cm)

plt.figure(figsize=(8,6))
sns.heatmap(cm, annot=True, fmt="d", cmap="Blues")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")
plt.show()
```

Figure 1



CHAT

TERMINAL

```
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning$
.14 /home/ben/Documents/Development/MachineLearning/Lab_5/WilliamsBe
xercise2.py
  File "/home/ben/.local/lib/python3.14/site-packages/pandas/io/common.py", line 92
6, in get_handle
    handle = open(
        handle,
    ...<3 lines>...
        newline="",
    )
FileNotFoundError: [Errno 2] No such file or directory: 'fashion-mnist_train.csv'
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning$ cd Lab_5
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning/Lab_5$ cs Willi
amsBenjamin_Lab5/
Command 'cs' not found, but can be installed with:
sudo apt install csound
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning/Lab_5$ cd Willi
amsBenjamin_Lab5/
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning/Lab_5/WilliamsB
enjamin_Lab5$ cd Exercise2
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning/Lab_5/WilliamsB
enjamin_Lab5/Exercise2$ /usr/local/bin/python3.14 /home/ben/Documents/Development/M
achineLearning/Lab_5/WilliamsBenjamin_Lab5/Exercise2/exercise2.py
Prediction for trousers.bmp: 1
Prediction for bag.jpg: 8
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning/Lab_5/WilliamsB
enjamin_Lab5/Exercise2$ /usr/local/bin/python3.14 /home/ben/Documents/Development/M
achineLearning/Lab_5/WilliamsBenjamin_Lab5/Exercise1/exercise1.py
Data loaded successfully.
Training the model...
Model trained successfully.
Accuracy: 0.8559

Classification Report:
              precision    recall  f1-score   support

           0       0.80      0.82      0.81      1000
           1       0.96      0.97      0.97      1000
           2       0.78      0.76      0.77      1000
           3       0.87      0.88      0.87      1000
           4       0.78      0.81      0.79      1000
           5       0.93      0.92      0.93      1000
           6       0.64      0.60      0.62      1000
           7       0.91      0.92      0.91      1000
           8       0.95      0.95      0.95      1000
           9       0.93      0.94      0.94      1000

    accuracy                           0.86     10000
   macro avg       0.85      0.86      0.86     10000
weighted avg       0.85      0.86      0.86     10000


Confusion Matrix (Text):
[[821   3  14  39   0   3 104   0  16   0]
 [  1 974   1  15   0   2   7   0   0   0]
 [ 17   3 758  12 109   0  93   0   8   0]
 [ 33  24  15 875  23   1  25   0   4   0]
 [  1   2  81  28 805   0  80   0   3   0]
 [  0   4   0   0   0 921   1  47   4  23]
 [154   5 103  32  92   0 598   0  16   0]
 [  0   0   0   0   0  37   0 918   0  45]
 [  5   0   5   5   5   6  20   4 948   2]
 [  0   0   0   0   0  16   0  42   1 941]]
```

CHAT

Ln 21, Col 68 Spaces: 4 UTF-8 LF {} Python Python 3.14.3

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TIMELINE

exercise1.py U | image.png U | fashion-mnist_train.csv U | exercise2.py U

Lab_5 > WilliamsBenjamin_Lab5 > Exercise2 > exercise2.py > ...

```python
import numpy as np
import pandas as pd
import cv2
from sklearn.linear_model import LogisticRegression

train_df = pd.read_csv("fashion-mnist_train.csv")
test_df = pd.read_csv("fashion-mnist_test.csv")

y_train = train_df.iloc[:, 0].to_numpy()
X_train = train_df.iloc[:, 1:].to_numpy() / 255.0

def preprocess_image(path):
    img = cv2.imread(path, cv2.IMREAD_GRAYSCALE)
    img = img / 255.0
    return img.reshape(1, -1)

img1 = preprocess_image("trousers.bmp")
img2 = preprocess_image("bag.jpg")

model = LogisticRegression(solver='saga', max_iter=1000, tol = 1e-2)
model.fit(X_train, y_train)

pred1 = model.predict(img1)[0]
pred2 = model.predict(img2)[0]

print("Prediction for trousers.bmp:", pred1)
print("Prediction for bag.jpg:", pred2)
```

CHAT

TERMINAL — Python - Exercise2

```
/usr/local/bin/python3.14 /home/ben/Documents/Development/MachineLearning/Lab_5/WilliamsBenjamin_Lab5/Exercise2/exercise2.py
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning$ /usr/local/bin/python3.14 /home/ben/Documents/Development/MachineLearning/Lab_5/WilliamsBenjamin_Lab5/Exercise2/exercise2.py
Traceback (most recent call last):
  File "/home/ben/Documents/Development/MachineLearning/Lab_5/WilliamsBenjamin_Lab5/Exercise2/exercise2.py", line 6, in <module>
    train_df = pd.read_csv("fashion-mnist_train.csv")
  File "/home/ben/.local/lib/python3.14/site-packages/pandas/io/parsers/readers.py", line 873, in read_csv
    return _read(filepath_or_buffer, kwds)
  File "/home/ben/.local/lib/python3.14/site-packages/pandas/io/parsers/readers.py", line 300, in _read
    parser = TextFileReader(filepath_or_buffer, **kwds)
  File "/home/ben/.local/lib/python3.14/site-packages/pandas/io/parsers/readers.py", line 1645, in __init__
    self._engine = self._make_engine(f, self.engine)
                   ~~~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^
  File "/home/ben/.local/lib/python3.14/site-packages/pandas/io/parsers/readers.py", line 1904, in _make_engine
    self.handles = get_handle(
                   ~~~~~~~~~~^
        f,
        ^^
    ...<6 lines>...
        storage_options=self.options.get("storage_options", None),
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
    )
    ^
  File "/home/ben/.local/lib/python3.14/site-packages/pandas/io/common.py", line 926, in get_handle
    handle = open(
        handle,
    ...<3 lines>...
        newline="",
    )
FileNotFoundError: [Errno 2] No such file or directory: 'fashion-mnist_train.csv'
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning$ cd Lab_5
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning/Lab_5$ cs WilliamsBenjamin_Lab5/
Command 'cs' not found, but can be installed with:
sudo apt install csound
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning/Lab_5$ cd WilliamsBenjamin_Lab5/
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning/Lab_5/WilliamsBenjamin_Lab5$ cd Exercise2
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning/Lab_5/WilliamsBenjamin_Lab5/Exercise2$ /usr/local/bin/python3.14 /home/ben/Documents/Development/MachineLearning/Lab_5/WilliamsBenjamin_Lab5/Exercise2/exercise2.py
Prediction for trousers.bmp: 1
Prediction for bag.jpg: 8
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning/Lab_5/WilliamsBenjamin_Lab5/Exercise2$
```

CHAT

Ln 27, Col 40 | Spaces: 4 | UTF-8 | LF | Python | Python 3.14.3